

① Ejercicio 1

Ej. 4.1

Tipo de punta (TRATAMIENTO)	Prueba (bloques)			Σ	a(c) = 3 b(f) = 3 N = 9
	1	2	3		
1	9.3	9.4	9.6	28.3	
2	9.4	9.3	9.8	28.5	
3	9.2	9.4	9.5	28.1	
Σ	27.9	28.1	28.9	(84.9)	

$$SCE = STC - SCT - SCB$$

$$C = \frac{(84.9)^2}{9} = 800.89$$

$$STC = (9.3^2 + 9.4^2 + 9.2^2 + 9.4^2 + 9.3^2 + 9.4^2 + 9.6^2 + 9.8^2 + 9.5^2) - 800.89 = 0.0260$$

$$SCT = \left(\frac{28.3^2}{3} + \frac{28.5^2}{3} + \frac{28.1^2}{3} \right) - 800.89 = 0.027$$

$$SCB = \left(\frac{27.9^2}{3} + \frac{28.1^2}{3} + \frac{28.9^2}{3} \right) - 800.89 = 0.187$$

$$SCE = 0.0260 - 0.027 - 0.187 = 0.047$$

Encuentre Ftablas para los bloques.

Fuente de Variación	Suma de cuadrados	Grados de libertad	Cuadrado medio	f_c
Tratamiento	0.027	2	0.013	1.143
Bloque	0.187	2	0.093	8.0
Error	0.047	4	0.012	
Total	0.260	8	X	

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$$\alpha = 0.05 = 5\% \text{ - Grado de confianza}$$

Pruebas de hipótesis

$$H_0 = \mu_1 = \mu_2 = \mu_3$$

H_a = Al menos 1 media es diferente

Grados libertad

$$G_T = a - 1$$

$$G_B = b - 1$$

$$G_E = (a-1)(b-1)$$

$$G_{\text{total}} = N - 1$$

$$f_c > f_T = \text{Se rechaza } H_0$$

$$f_c < f_T = \text{No se rechaza } H_0$$

$$f_T(2, 4)$$

$$f_c \quad 1.143 < f_T \quad 6.94$$

Existe la evidencia de la igualdad de las medias

$$\sqrt{550.0} = 23.24 \quad \text{P8.008} - \left(\frac{1.82}{5} + \frac{2.82}{5} + \frac{3.82}{5} \right) = 73.2$$

$$\sqrt{581.0} = 24.1 \quad \text{P8.008} - \left(\frac{1.82}{5} + \frac{1.82}{5} + \frac{1.82}{5} \right) = 83.2$$

$$\sqrt{590.0} = 24.3 \quad \text{P8.008} - \left(\frac{1.82}{5} + \frac{1.82}{5} + \frac{1.82}{5} \right) = 83.2$$

	Grupos	Media	Desviación	Grados libertad
1	510.0	2	550.0	2
2	580.0	2	581.0	2
3	510.0	2	590.0	2
		8	550.0	8

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② Ejercicio 2

Prueba (bloques)

Tipo de punto
(tratamiento)

$$a(c) = 3$$

$$b(f) = 4$$

$$N = 12$$

	1	2	3	Σ
1	9.3	9.4	9.6	28.3
2	9.4	9.3	9.8	28.5
3	9.2	9.4	9.5	28.1
4	9.7	9.6	10.0	29.3
Σ	37.6	37.7	38.9	114.2

$$SCE = STC - SCT - SCB = -90.527$$

$$C = \frac{(114.2)^2}{12} = 1,086.803$$

$$STC = (9.3^2 + 9.4^2 + 9.2^2 + 9.7^2 + 9.4^2 + 9.3^2 + 9.4^2 + 9.6^2 + 9.6^2 + 9.8^2 + 9.5^2 + 10^2) - 1,086.803 = 0.597$$

$$SCT = \left(\frac{28.3^2}{4} + \frac{28.5^2}{4} + \frac{28.1^2}{4} + \frac{29.3^2}{4} \right) - 1,086.803 = -271.493$$

$$SCB = \left(\frac{37.6^2}{3} + \frac{37.7^2}{3} + \frac{38.9^2}{3} \right) - 1,086.803 = 362.617$$

Encuentre Ftablas para los bloques

Fuente de variación	Suma de cuadrados	Grados de libertad	Cuadrado medio	f_c
Tratamiento	-271.493	2	-135.746	9.0
Bloque	362.617	3	120.872	-8.01
Error	-90.527	6	-15.087	
Total	0.597	11	X	

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$$\alpha = 0.05 = \underline{95\%} \quad - \text{Grados de confianza}$$

Pruebas de Hipotesis

$$H_0 = \mu_1 = \mu_2 = \mu_3$$

$$H_a = \text{Al menos 1 media es diferente}$$

Grados libertad

$$G_T = a - 1$$

$$G_B = b - 1$$

$$G_E = (a - 1)(b - 1)$$

$$G_{Tot} = N - 1$$

$$f_c > f_t = \text{Se rechaza } H_0$$

$$f_c < f_t = \text{No se rechaza } H_0$$

$$f_t(2, 6)$$

$$f_c \ 9.0 > f_t \ 5.14$$

No existe evidencia
de la igualdad de las
medias

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③ Ejercicio 3

Pruebas (bloques)

Tipo de punta
(Tratamiento)

$a(c) = 4$

$b(f) = 4$

$N = 16$

	1	2	3	4	Σ
1	9.3	9.4	9.6	10.0	38.3
2	9.4	9.5	9.8	9.9	38.6
3	9.2	9.4	9.5	9.7	37.8
4	9.7	9.6	10.0	10.2	39.5
Σ	37.6	37.9	38.9	39.8	154.2

$$SCE = STC - SCT - SCB = 0.062$$

$$C = \frac{(154.2)^2}{16} = 1,486.102$$

$$STC = (9.3^2 + 9.4^2 + 9.2^2 + 9.7^2 + 9.4^2 + 9.5^2 + 9.6^2 + 9.8^2 + 9.5^2 + 10.0^2 + 10.0^2 + 9.9^2 + 9.7^2 + 10.2^2) - 1,486.102 = 1.198$$

$$SCT = \left(\frac{38.3^2}{4} + \frac{38.6^2}{4} + \frac{37.8^2}{4} + \frac{39.5^2}{4} \right) - 1,486.102 = 0.383$$

$$SCB = \left(\frac{37.6^2}{4} + \frac{37.9^2}{4} + \frac{38.9^2}{4} + \frac{39.8^2}{4} \right) - 1,486.102 = 0.753$$

Encuentre FIABLAS para los bloques

Fuente de variación	suma de cuadrados	Grado Libertad	Cuadrado medio	f_c
Tratamiento	0.383	3	0.127	18.532
Bloque	0.753	3	0.251	36.435
Error	0.062	9	0.006	
Total	1.198	15	X	

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$\alpha = 0.05 = 95\%$ - Grado de confianza

Pruebas de Hipotesis

$$H_0 = \mu_1 = \mu_2 = \mu_3$$

$H_a =$ Al menos 1 media es diferente

Grado libertad

$$G_T = a - 1$$

$$G_B = b - 1$$

$$G_E = (a - 1)(b - 1)$$

$$G_{Tot} = N - 1$$

$f_c > f_T =$ Se rechaza H_0

$f_c < f_T =$ No se rechaza H_0

$$f_T = (3, 9)$$

$$f_c 18.532 > f_T 3.86$$

No existe evidencia
de la igualdad de las
medias